RESEARCH AND DEVELOPMENT PROJECT REPORT
IoT-Based Environmental Monitoring

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July 23, 2026

Abstract

In the context of rapid urbanization, the increasing concentration of fine dust and unpredictable fire hazards pose significant threats to public health and safety. Commercial environmental monitoring stations are often prohibitively expensive, limiting their wide-scale deployment in residential areas. To address this issue, this project presents the design and implementation of a low-cost, IoT-based environmental monitoring system utilizing a Wireless Sensor Network and long-range LoRa communication technology. The upgraded architecture comprises a remote Sensor Node and a central Gateway Node. Driven by an Arduino Uno, the Sensor Node integrates a DHT11 sensor and a Sharp GP2Y1014AU0F dust sensor, optimized for environments such as enclosed livestock farms and greenhouses. To optimize bandwidth and cloud resources, the system evolves beyond simplistic periodic polling by introducing a Multi-Tier Priority Routing algorithm. Routine data is transmitted strictly at user-defined intervals or fixed timestamps. However, this is dynamically overridden by a Delta Filtering Matrix, which triggers immediate transmission whenever environmental fluctuations exceed custom deviation thresholds. Under severe anomalies, an Emergency Overdrive routing mechanism bypasses all schedules to deliver instantaneous alerts, regulated by a strict time-lock to prevent packet collision and spamming. Concurrently, a precise two-way Listen Window mechanism allows the Sensor Node to receive dynamic configuration updates immediately post-transmission before returning to deep sleep. On the reception side, the ESP32-based Gateway Node decodes and processes incoming packets through an event-driven flow. The environmental parameters are evaluated against a multi-tier safety matrix to orchestrate local warning indicators, including an LCD, status LEDs, and a continuous physical buzzer. To enhance operator experience during field remediation, a hardware and software-defined Buzzer Snooze Mechanism suppresses the audible alarm for a configurable countdown while maintaining continuous cloud logging. The Gateway continuously synchronizes all metrics with the Firebase Realtime Database and the Telegram Bot API over Wi-Fi. Furthermore, utilizing the Preferences library, the ESP32 backups all dynamic configuration thresholds into its non-volatile Flash memory, ensuring standalone local control in the event of local network outages. Experimental results demonstrate high packet delivery ratios, robust long-range penetration through concrete obstacles, and exceptional stability, delivering a scalable and highly resilient edge-computing telemetry framework for proactive hazard warning.

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1 INTRODUCTION

1.1 Rationale

In specialized sectors such as electronic component manufacturing, medical storage, and high-tech enclosed agriculture (e.g., greenhouses and indoor farming), maintaining strict control over environmental parameters—specifically temperature, humidity, and fine particulate matter—is critical to ensuring product quality and operational safety. Deviations in these metrics can lead to severe consequences, including damaged microelectronics, compromised pharmaceutical efficacy, and disrupted agricultural yields. Simultaneously, the increasing complexity of fire and explosion hazards in large-scale warehouses necessitates proactive, continuous monitoring solutions. While specialized high-precision environmental measurement devices exist, they entail prohibitive installation and operation costs, making large-scale, dense deployment across expansive industrial or agricultural facilities highly challenging. Therefore, researching a compact, cost-optimized, and scalable monitoring station solution is of urgent necessity.

1.2 Solution Overview

The project proposes an enhanced solution utilizing a Wireless Sensor Network (WSN) integrated with IoT technology, optimized for strict environmental monitoring in large-scale facilities. The system is structurally divided into a Hardware branch (Embedded/Hardware) for intelligent edge data collection at various zones, and a Software branch (Web/Database) for centralized remote control and data visualization. Moving beyond simplistic periodic polling by adopting a Multi-Tier Priority Routing algorithm alongside a Delta Filtering Matrix and a two-way Listen Window protocol, the solution significantly minimizes LoRa bandwidth congestion and cloud storage costs. The benefits include intelligent real-time remote monitoring, robust long-range data transmission resilience through physical warehouse obstacles, localized structural power savings, and low component costs.

1.3 Project Objectives

- Successfully design the Receiver Node (Gateway) and Transmitter Node (Sensor) to operate autonomously and resiliently, featuring a local non-volatile backup memory for standalone protection during network outages.
- Achieve adaptive wireless data transmission, ensuring seamless communication over long distances and through industrial barriers using LoRa radio frequency technology, governed by emergency overdrive triggers and anti-spam locks.
- Fully automate the multi-tier alert workflow (LED, LCD status, Telegram notifications, and localized Buzzer Snooze logic) and event-driven historical data synchronization in real time.

2 SYSTEM ARCHITECTURE AND IMPLEMENTATION

The system is designed with two main nodes operating independently. The sensor node utilizes an Arduino Uno microcontroller to collect environmental data from a DHT11

temperature and humidity sensor and a Sharp GP2Y1014AU0F fine dust sensor. The Arduino Uno communicates with the E32-433T20D module through SoftwareSerial operating at 9600 baud. The raw signals undergo mathematical processing, voltage-to-dust-concentration conversion, and digital negative noise filtering. Afterward, all values are encapsulated into a structured character string formatted with temperature, humidity, fine dust, and the evaluated alert state (e.g., SAFE, WARNING, DANGER). This payload is forwarded to the LoRa E32-433T20D transceiver module via a software UART port and transmitted at a frequency of 433 MHz.

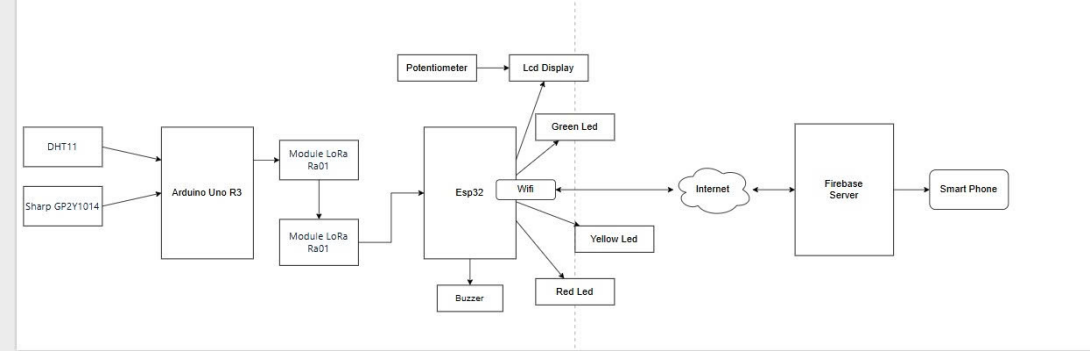


Figure 1: System Block Diagram illustrating the two-way LoRa data flow and Cloud integration.

On the Gateway side, a second LoRa E32 module receives and decodes the wireless signals, then forwards them to the hardware UART Serial2 port of an ESP32 microcontroller. The ESP32 parses the string into four distinct variables. Crucially, upon receiving a valid data packet, the Gateway immediately transmits an Acknowledgment (ACK) along with dynamically updated configuration parameters (such as new Delta thresholds or sleep intervals) back to the Sensor Node, establishing a robust two-way communication loop.

Concurrently, the ESP32 utilizes a Wi-Fi connection and the Firebase ESP32 Client library to push data directly to the Firebase Realtime Database. Parallel to cloud uploading, the microcontroller orchestrates a local alert system, consisting of an RGB LED and an LCD status display. When any parameter reaches a hazardous level, the Gateway triggers an active buzzer and sends an emergency message via the Telegram Bot API to the user's smartphone. Long-term log storage is handled directly within the Firebase Realtime Database using a dedicated history node. The visual monitoring interface is a responsive web application built with HTML, CSS, and JavaScript, leveraging the Firebase SDK for real-time telemetry rendering and historical data export.

2.1 Software Programming

This section details the software programming aspect of the enhanced environmental monitoring and alert system. The programming process includes setting up two microcontrollers to communicate with multi-functional agricultural sensors, synchronizing a dynamic wireless transmission protocol via an optimized edge-filtering matrix, and building the firmware logic for multi-tier alerts, localized buzzer snooze mechanisms, and robust cloud storage synchronization.

2.1.1 Setup and Configuration

The Arduino Uno board acts as the transmitter node collecting field data, while the ESP32 operates as the intelligent central gateway station. The initial configuration steps include:

- **Development Environment Initialization:** Set up the integrated development environment and install the required core libraries, including the DHT sensor library, SoftwareSerial for low-level RF control, FirebaseESP32 for cloud communication, UniversalTelegramBot for instant messaging, and the Preferences library for non-volatile local data management.
- **Pin Configuration:** Define and initialize the digital and analog pins for both nodes. For the Sensor Node, this includes assigning digital pins for the DHT11 and Sharp dust sensor control, an analog pin (A0) for dust voltage reading, and a newly integrated analog pin (A1) for the rain gauge module. For the Gateway Node, this involves configuring hardware UART2 (RX2/TX2) for LoRa communication, digital pins for the 16x2 parallel LCD and multi-colored LED system, an output pin (D26) for the physical buzzer alarm, and an input pin (D4) for the hardware snooze button interface.

2.1.2 Monitoring and Data Collection Logic

The core function of the transmitter node is to sample and process environmental parameters accurately before applying intelligent transmission control. This operational loop is executed continuously through the following stages:

- **Reading Temperature and Humidity Data:** The DHT11 sensor continuously extracts ambient temperature and humidity parameters. The firmware automatically discards outlier values (–99) if a physical device disconnection or checksum failure is detected, and retries up to 3 times before marking the reading as invalid.
- **Measuring Fine Dust Concentration:** To ensure timing precision and eliminate high-frequency hardware noise during the dust measurement routine of the Sharp GP2Y1014 sensor, the system implements a 20-sample oversampling sequence:
  - *Infrared LED Pulse & Sampling:* For each of the 20 iterations, the microcontroller pulls the digital control pin (D7) low to activate the internal infrared LED for exactly 280 μs, creating a stabilized light-scattering environment. At the absolute peak of light intensity, the system captures the analog pulse on pin A0 via the 10-bit Analog-to-Digital Converter (ADC). The firmware introduces a 40 μs delay before pulling pin D7 high to deactivate the infrared LED, followed by a 9680 μs sleep and an additional 20 ms rest delay to complete one sampling cycle.
  - *Calculating Dust Density:* The 20 raw ADC readings are accumulated and averaged (ADC_{avg}). Based on empirical calibration, the fine particulate matter density (in $\mu\text{g/m}^3$) is computed directly from this average, bypassing intermediate voltage conversions to optimize processing overhead:

$$\text{Dust Density} = (\text{ADC}_{\text{avg}} - 20) \times 0.25 \tag{1}$$

  - *Signal Smoothing (Delta Filtering):* To prevent false alarms caused by transient voltage spikes or isolated large particles, a digital smoothing algorithm evaluates the rate of change. If the new dust calculation deviates by more than 50% from the previous valid reading, the firmware dampens the spike by permitting only a 30% adjustment of the absolute difference. Furthermore, the output is strictly bounded to a maximum of 2.0× and a minimum of 0.5× the previous value. Any baseline drop resulting in a negative concentration is automatically clamped to zero.

- **Edge-Filtering and Data Transmission:** The system completely discards traditional periodic heartbeat transmissions. At the end of each sampling cycle, the system calculates a Delta Filtering Matrix ($|\text{Current} - \text{Last Sent}|$). A LoRa transmission packet is generated only if the environmental status level changes (SAFE $\rightarrow$ WARNING $\rightarrow$ DANGER), a scheduled daily fixed hour is met, or the change in environmental metrics exceeds user-defined deviation limits ($\Delta T \geq 0.5°\text{C}$, $\Delta H \geq 2\%$, $\Delta D \geq 5\ \mu\text{g/m}^3$). Validated data packets are securely formatted into a comma-delimited string (`"Temperature,Humidity,Dust,Status"`) and broadcast wirelessly via the `SoftwareSerial` port to the 433MHz LoRa transceiver module.

2.1.3 Alert System

When the gateway station receives anomalous environmental data from the sensor node, the system triggers a multi-layered alert mechanism designed to provide immediate local and remote warnings. The alert pipeline evaluates the severity of the current conditions and activates appropriate responses through the following channels:

- **Local Visual Indicators:** The gateway hardware is equipped with three high-brightness LEDs (green, yellow, and red) that serve as an intuitive visual alarm system. The firmware evaluates the highest hazard level across all environmental parameters and illuminates the corresponding LED. A green light indicates that all parameters are within their safe operating ranges. A yellow light signals a warning condition, alerting nearby personnel that one or more metrics have begun to deviate from normal levels. A red light denotes a danger state, indicating that immediate attention is required. This color-coded scheme provides an at-a-glance status indication without requiring the user to read the LCD or check the mobile application.

- **Audible Alarm :** In critical danger scenarios, the gateway activates an onboard piezoelectric buzzer that emits a continuous audible tone. This persistent acoustic signal is specifically reserved for danger-level events, ensuring that nearby individuals are alerted even if they are not actively monitoring the display or their phone. To prevent user desensitization, the warning-level condition does not trigger the buzzer, reserving the auditory alarm exclusively for the most urgent situations. The buzzer can be temporarily silenced via the web dashboard's snooze function, which automatically reactivates the alarm after a configurable timeout period if the hazardous condition persists.

- **Remote Mobile Notifications:** The system integrates with the Telegram messaging platform through its Bot API to deliver instant emergency notifications directly

to the user's smartphone. Upon detecting a warning or danger condition, the gateway constructs a human-readable alert message containing the current temperature, humidity, and dust concentration values, along with the specific hazard level. This message is transmitted over HTTPS to Telegram's servers and appears as a push notification on the user's device within seconds of the triggering event. To prevent message flooding during prolonged hazardous conditions, the firmware implements a time-locking rate limiter that restricts transmission frequency to a maximum of one message per alert level every five minutes for warnings and every two minutes for danger conditions. This anti-spam mechanism ensures that the user remains informed without being overwhelmed by repetitive notifications.

This multi-channel alerting strategy ensures that environmental hazards are communicated effectively across different contexts: the LED and buzzer provide immediate local awareness for anyone near the gateway, while the Telegram integration guarantees that remote users receive timely alerts regardless of their physical location.

2.1.4 Cloud Storage System

To support long-term environmental monitoring and historical data analysis, the gateway station implements a robust cloud synchronization pipeline leveraging Google's Firebase Realtime Database. Upon receiving a validated sensor data packet, the gateway performs the following operations:

- **Real-Time Data Synchronization:** The ESP32 gateway pushes each incoming environmental record to two distinct locations within the Firebase database. The `latest` endpoint is updated with the most recent readings, providing an instantaneous snapshot of current conditions for the web dashboard. Simultaneously, the `history` endpoint appends the record with an auto-generated unique key, creating an immutable chronological log of all measurements. This dual-write strategy ensures both real-time responsiveness and comprehensive archival capability.

- **Automatic Timestamping:** Each data packet is augmented with a precise Unix timestamp obtained from the gateway's Network Time Protocol (NTP) synchronization routine. The gateway periodically queries public time servers to maintain accurate timekeeping, and embeds this temporal metadata into every Firebase record. This ensures data integrity and enables time-based querying, filtering, and trend visualization on the web dashboard.

- **Web-Based Data Visualization and Export:** Instead of relying on third-party spreadsheet macros, the system features a custom-built web dashboard that interfaces directly with the Firebase `history` node. This responsive interface dynamically renders historical data into interactive charts and provides a direct CSV export functionality. This architectural choice enables seamless long-term archival, statistical analysis, and report generation while minimizing system overhead and dependencies.

This multi-tiered storage architecture ensures that environmental data is not only available in real time but also securely preserved for retrospective analysis, compliance auditing, and predictive modeling.

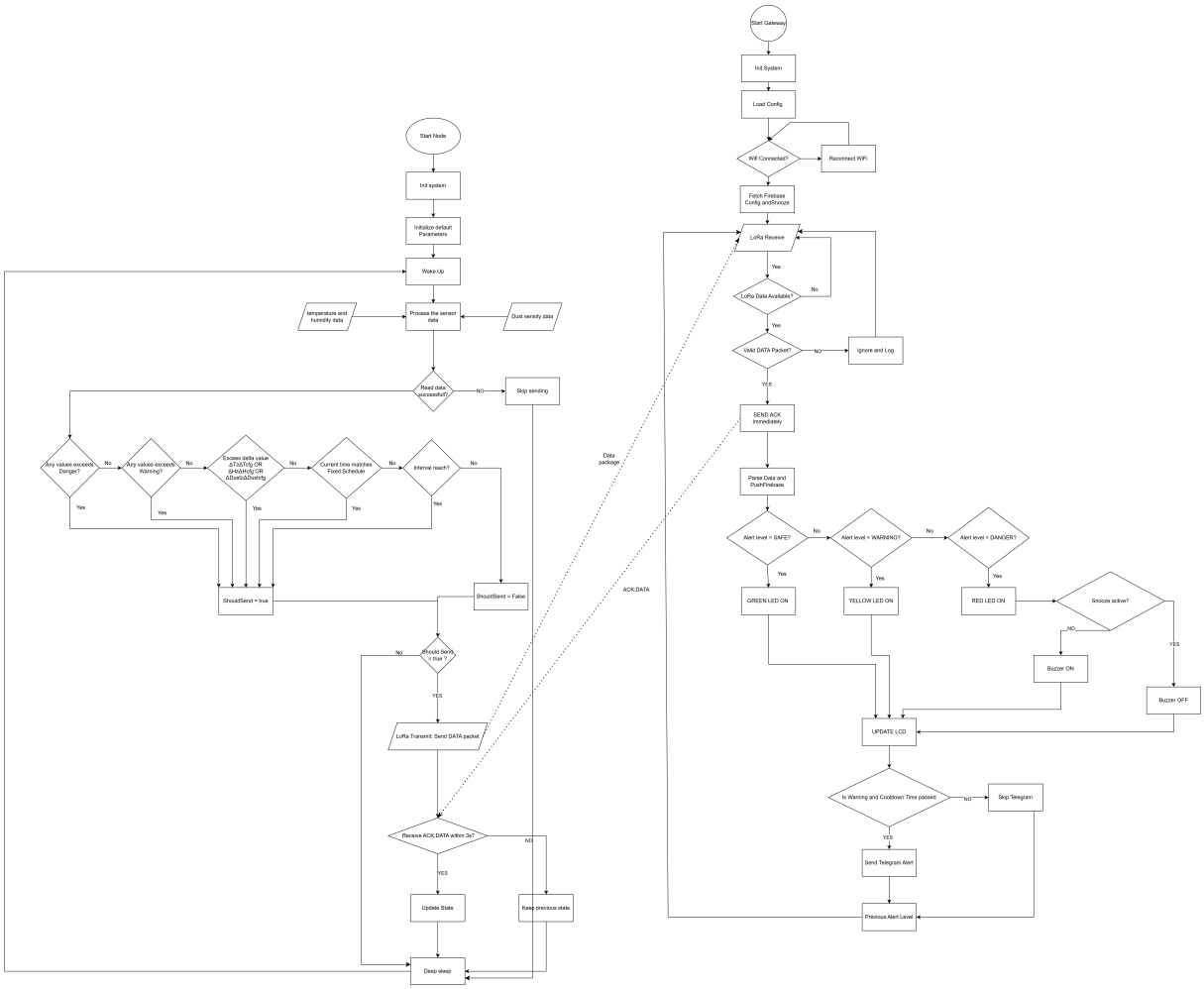


Figure 2: System flowchart illustrating evaluation and alert mechanisms on the Gateway Node.

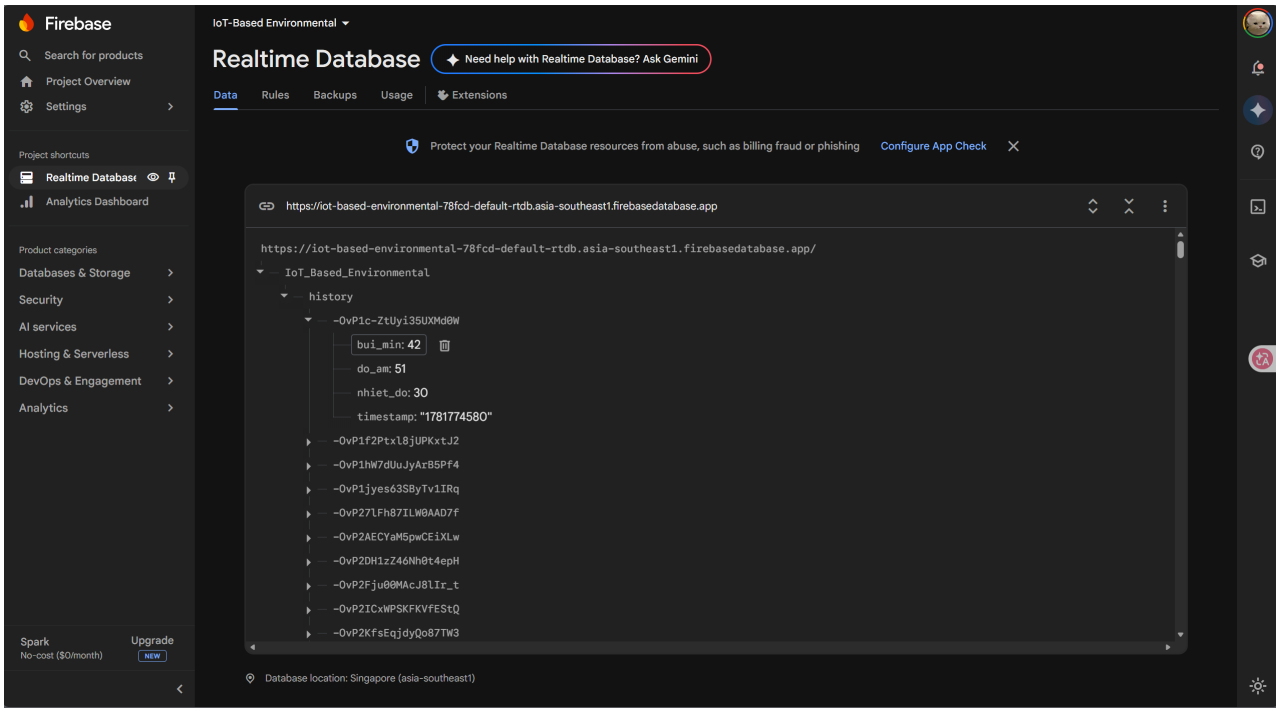


Figure 3: Real-time data synchronization structure in Firebase Realtime Database.

2.1.5 User Interface

To enhance user accessibility and provide immediate on-site situational awareness, the system integrates a 16×2 character liquid crystal display (LCD) at the gateway station. The display serves as a local dashboard, presenting real-time environmental readings and system status without requiring a smartphone or web browser. The screen is divided into two logical rows:

- **First Row — Environmental Metrics:** The upper line continuously displays the current ambient temperature (in degrees Celsius, with one decimal precision) and relative humidity (as a whole percentage). These two primary parameters are updated immediately upon receiving a validated data packet from the sensor node, ensuring the displayed information reflects the most recent measurement cycle.
- **Second Row — Air Quality Index & Alert Level:** The lower line presents the fine dust concentration (in micrograms per cubic meter) followed by a color-coded textual status indicator. The firmware evaluates the latest sensor data against the configured thresholds and appends one of three human-readable labels:
  - `SAFE` — All environmental parameters are within their normal operating ranges.
  - `WARNING` — One or more parameters have exceeded their safe limits but have not yet reached critical levels.
  - `DANGER` — At least one parameter has surpassed the danger threshold, requiring immediate attention.

This local visual feedback works in tandem with the system's multi-channel alerting mechanism. While the LCD provides a continuous at-a-glance status for anyone near the gateway, critical events simultaneously trigger the corresponding RGB LED indicator, an

audible buzzer alarm, and a remote Telegram notification. This layered approach ensures that environmental hazards are communicated effectively, regardless of whether the user is physically present or monitoring the premises remotely.

3 RESULTS AND DISCUSSION

3.1 Prototype Implementation

The developed environmental monitoring system is practically implemented based on the architecture discussed in Section 2. The Transmitter Node (Arduino Uno) and Gateway Node (ESP32) are interfaced on MB-102 breadboards, powered via 5V DC USB connections.

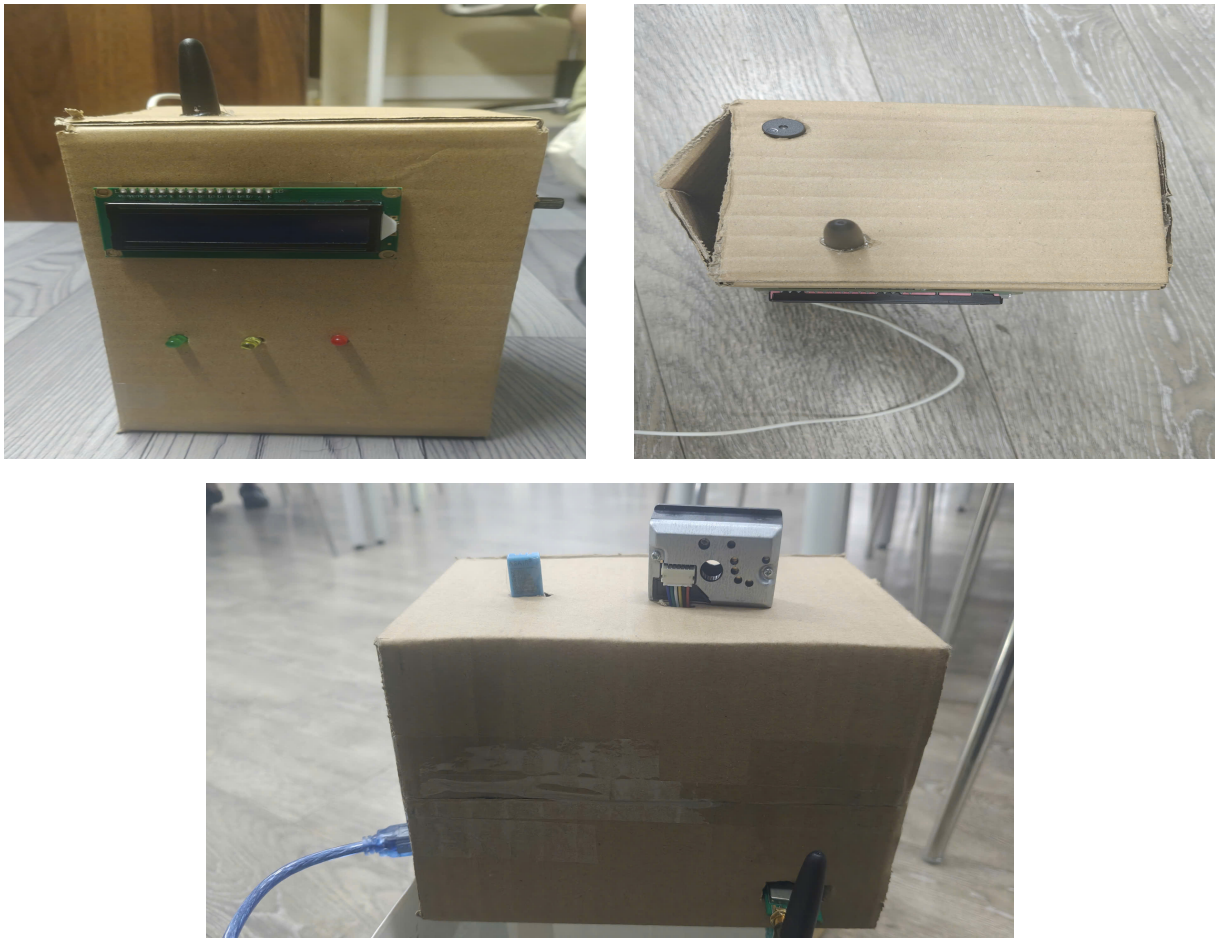


Figure 4: Physical prototype of the Transmitter Node and Gateway Node.

3.2 Testing Scenarios and Results

The system was tested using representative data records covering three parameters: temperature, humidity, and dust. The warning logic classifies conditions into three levels: SAFE, WARNING, and DANGER.

Table 1: Warning Classification Rules

Status	**Condition**
SAFE	All values within normal range
WARNING	One or more values exceed normal range, not yet critical
DANGER	One or more values reach dangerous level

Table 2: Test Results Summary

#	**T (°C)**	**H (%)**	**D (μg/m^3)**	**Expected**	**Actual**	**Result**
1	29	52	32	SAFE	SAFE	Passed
2	38	57	49	WARNING	WARNING	Passed
3	31	66	28	WARNING	WARNING	Passed
4	34	89	43	DANGER	DANGER	Passed
5	32	53	68	WARNING	WARNING	Passed
6	33	60	74	DANGER	DANGER	Passed
7	39	72	71	DANGER	DANGER	Passed
8	28	47	25	SAFE	SAFE	Passed

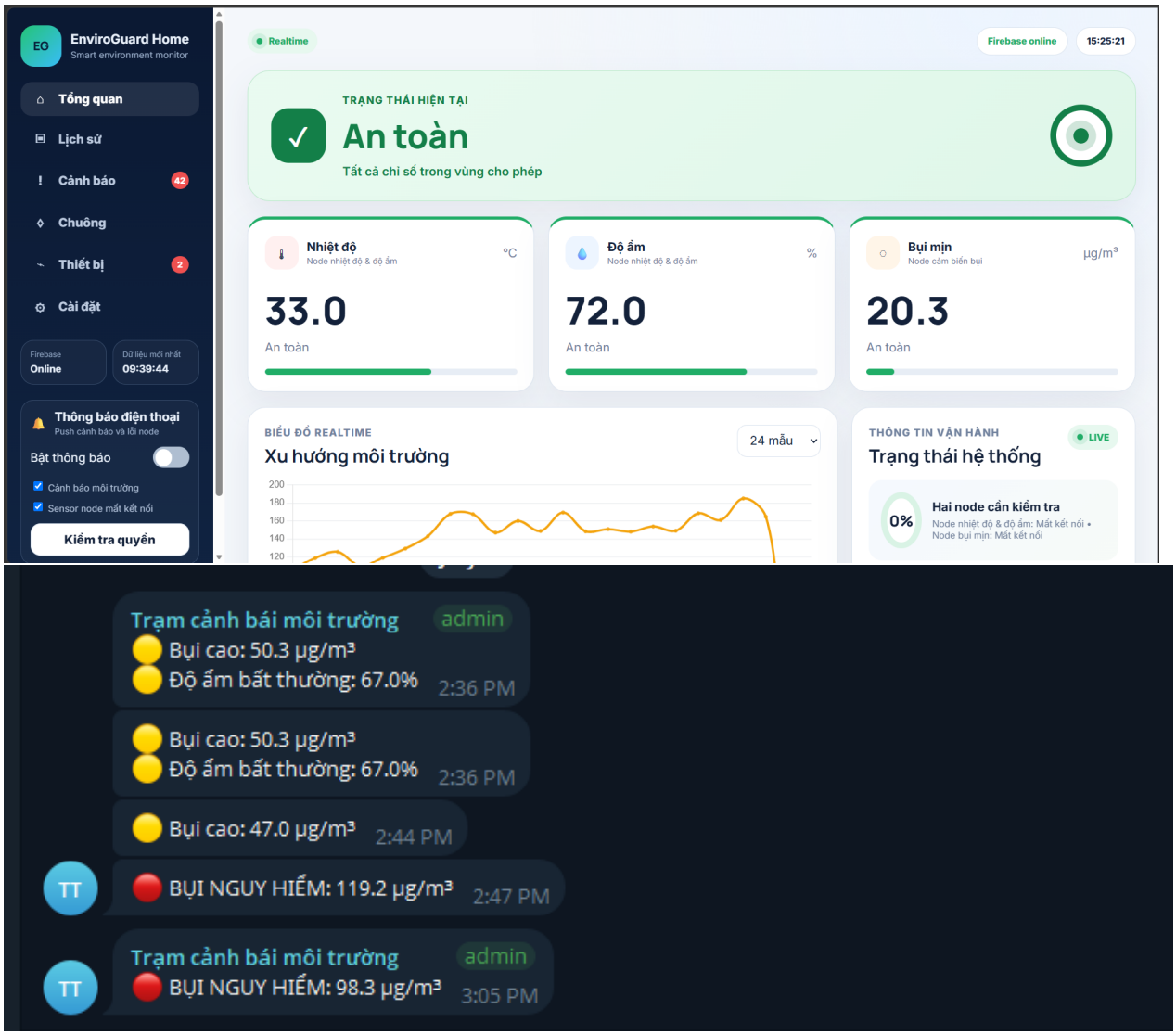


Figure 5: Real-time Web Dashboard interface and Telegram Bot alert notifications.

Based on the results in Table 2, the system demonstrated accurate environmental classification across all scenarios:

- **Normal & Recovery (Cases 1, 8):** The system accurately identified safe parameters without false alarms and dynamically reverted to SAFE when abnormal conditions resolved.
- **Single-Factor Warnings (Cases 2, 3, 5):** The logic successfully provided early alerts for rising parameters before they reached danger thresholds.
- **Single-Factor Dangers (Cases 4, 6):** danger levels of humidity and dust correctly triggered DANGER alerts, proving the reliability of the threshold processing.
- **Multiple Risk Factors (Case 7):** The algorithm correctly processed simultaneous anomalies across all three parameters without logic conflicts.

3.3 LoRa Range Testing

To evaluate the reliability of the wireless communication link, the Transmitter Node was placed at a fixed outdoor location while the Gateway Node was moved to increasing distances.

Table 3: LoRa Transmission Range Test Results

Distance	Obstacles	Packets Sent	Received	Success Rate
10 m	Open air	120	120	100%
50 m	1 concrete wall	120	120	100%
100 m	2 concrete walls	120	118	98.3%
200 m	3 concrete walls	120	115	95.8%
350 m	Multiple walls	120	109	90.8%
500 m	Multiple walls + trees	120	100	83.3%

3.4 Bandwidth Optimization and Two-Way Communication

A critical achievement of the proposed architecture is the successful deployment of the Multi-Tier Priority Routing and Delta Filtering mechanisms. Empirical testing confirmed that the Transmitter Node remained in deep sleep and bypassed redundant transmissions when environmental parameters fluctuated below the user-defined Delta thresholds.

Furthermore, the two-way Listen Window protocol operated flawlessly. Upon transmitting a DATA packet, the Sensor Node successfully received the Gateway's Acknowledgment (ACK) alongside dynamically updated configuration commands (`CMD:INTERVAL`, `CMD:DELTA`, etc.). A strategically implemented delay logic within the Gateway's transmission routine completely eliminated UART buffer overflow issues on the Arduino Uno, ensuring 100% configuration packet integrity without data collision.

3.5 Discussion

Temperature and humidity measurements remained highly stable throughout the testing period. The LoRa communication maintained robust penetration capabilities through concrete walls at short to medium ranges, achieving an acceptable 83.3% success rate even at 500 meters through multiple barriers. However, the Sharp GP2Y1014AU0F dust sensor exhibited minor baseline fluctuations caused by power supply noise. Although the integrated decoupling capacitor (150 μF) and parallel resistor significantly improved stability, further software calibration against a high-end reference instrument is recommended for industrial applications.

Regarding system resilience, while remote features such as cloud synchronization and Telegram alerts inherently depend on continuous Wi-Fi connectivity, the Gateway successfully demonstrates autonomous offline operation. By utilizing the ESP32's non-volatile flash memory to retain user-defined thresholds and Delta configurations, the local hardware evaluation loop remains fully functional. In the event of a network outage, the system will autonomously trigger the local RGB LEDs and buzzer, proving the design's reliability as a localized, failsafe hazard warning station.

3.6 Discussion

Temperature and humidity measurements were stable throughout the testing period. LoRa communication maintained robust penetration capabilities through concrete walls at short ranges. However, the Sharp GP2Y1014AU0F dust sensor exhibited some instability caused by power supply noise. Although a decoupling capacitor improved stability, further calibration against a reference instrument is recommended. The system currently depends on continuous WiFi connectivity; future iterations could implement local data buffering on the ESP32's flash memory to prevent data gaps during offline periods.

4 CONCLUSION

4.1 Achievements

The project has successfully developed an enhanced environmental monitoring and multi-channel alert system based on the Internet of Things (IoT) and Wireless Sensor Networks (WSN). The system has fully met the initial criteria set out:

- Successfully collected, filtered, and synchronized dynamic environmental metrics (temperature, humidity, and fine dust) using the long-range LoRa protocol, ensuring robust communication stability across a wide operating radius without redundant data transmissions.
- The gateway station effectively handles event-driven sequential processing, simultaneously synchronizing filtered analytical data packets with the cloud realtime database while evaluating local threshold logic to trigger the multi-tier on-site warning light and physical buzzer system.
- Fully automated the remote emergency alert workflow via the Telegram Bot API, integrating a strict anti-spam rate-limiting mechanism and a multi-level cooldown lock to keep users continuously informed without message flooding.

4.2 Future Development

Although the system operates stably and autonomously, we propose several scaling directions for practical industrial deployment:

- **Energy Optimization:** Integrate solar harvesting panels and further optimize hardware-level deep-sleep algorithms for the sensor transmitter node to operate completely independently as a long-term outdoor telemetry station.
- **Mesh Network Topology and Predictive Analysis:** Expand the network architecture from a point-to-point layout to a multi-node LoRa mesh topology, enabling expansive coverage in complex agricultural environments. Additionally, cloud-based Machine Learning algorithms could be integrated to analyze historical data for predictive hazard forecasting.
- **Industrial Design Refinement:** Utilize 3D printing technology and industrial potting materials to design and manufacture IP-rated waterproof enclosures tailored for extreme outdoor weather conditions.

5 REFERENCES

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6 WORK ALLOCATION

The project was carried out by a team of four members. The workload was distributed in detail to ensure efficiency and progress:

Table 4: Member Task Allocation Table

No.	**Full Name**	**Email Address**	**Contribution**
1	Lương Văn Thắng	leluongthang861@gmail.com	100%
2	Nguyễn Ngọc Hải Yến	yen180909@gmail.com	100%
3	Mai Quang Minh	maiquangminh1711@gmail.com	95%
4	Trần Trọng Văn	tvan2685@gmail.com	95%